CDA: Post-hoc Domain Generalization by Seeking Distributional Canalization

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Abstract

Current post-hoc domain generalization methods are usually justified by parameter-space flatness, diversity among models being averaged, or plain validation performance. We study *distributional canalization*, namely source-mixture sensitivity, as a source-only post-hoc criterion for evaluating deployable models. Our central theoretical claim has two parts. First, under a source-supported hidden-shift model, target risk is upper bounded by mean source risk plus a source-mixture sensitivity term, so models with lower $\bar{L}(\theta) + \mathcal{S}_\rho(\theta)$ are more generalizable in that regime. Second, on a fixed retained checkpoint family, a direct deployed-soup upper bound yields a practical surrogate for CDA, built from worst-domain source fit, an interaction-based hidden-shift penalty, and a geometry-controlled mergeability term. The theorem-bearing contribution is organized around a model-level local tangent-space ambiguity set over source mixtures, not around soups themselves. We prove the target-risk bound, an exact closed form for the associated hidden-shift robust risk in the tangent model, variance and interaction decompositions that expose complementarity and redundancy, a sufficient low-curvature regime explaining why flatness works, and a constructive example showing why pooled flatness can miss source-mixture sensitivity. For the practical weighting problem on a fixed checkpoint family, we prove a direct retained-family observable upper bound for the deployed soup, convexity, and projected-subgradient convergence for the theorem-level surrogate. The analysis is explicitly conditional and local: it is a held-out-loss argument unless a separate empirical-to-population layer is added. New CDA empirical results remain intentionally marked **TBD**, while published benchmark context from prior post-hoc domain generalization papers is included to define the target bar.

1 Introduction

Post-hoc domain generalization matters because it is one of the few robustness strategies that can be attached to a strong empirical-risk-minimization pipeline without retraining. A single training run produces a dense checkpoint bank, and one deployable model is chosen only after training has already finished. This regime is operationally attractive and empirically credible. On DomainBed, SWAD reported an average accuracy of 66.9 across PACS, VLCS, OfficeHome, TerraIncognita, and DomainNet, improving ERM from 63.3 [4]. Later weight-averaging work pushed the frontier higher, including DIWA, whose strongest published linear-probe setting reports 68.0 average accuracy [12]. The practical question is therefore no longer whether post-hoc averaging can help. The unresolved question is *what criterion the final post-hoc model should optimize*.

The dominant explanation in this niche is flatness. SWAD states that finding flat minima reduces the domain generalization gap [4]. That theory is important, but it is incomplete in two ways. First, flatness mainly explains why averaging may be *safe*; it does not fully explain why some averaged solutions are more *useful* than others. Second, later work has emphasized that locality and diversity matter jointly under shift. DIWA explicitly frames weight averaging through a bias-variance-covariance-locality decomposition and argues that good OOD averaging requires a trade-off

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between diversity and locality [12]. These are not competing stories. They can be interpreted through a common source-side robustness lens.

This paper studies the following source-side post-hoc criterion:

> **Distributional canalization.** A model is preferable when its source performance is stable under small local redistributions of source support mass.

The claim is intentionally not specific to soups. It is a statement about models, weights, and minima. Soups enter only later, as one practical way to search for models with low source-mixture sensitivity using a completed checkpoint bank.

The intuition is direct. In post-hoc DG, the target domain is unavailable, but source data still contain latent heterogeneity. A future shift can be approximated locally as a nearby change in the mixture of source environments or source slices. A model is therefore desirable when its loss does not change much under those small redistributions. This object lives in *distribution space*, not parameter space. Flatness becomes a sufficient but not necessary mechanism: if nearby parameter perturbations barely alter the domain-loss vector, then source-mixture sensitivity is small, but the converse need not hold.

This perspective yields a two-step story. First, we show that under a source-supported hidden-shift assumption, target risk is controlled by mean source loss plus source-mixture sensitivity. Second, starting from a checkpoint bank, we derive a practical soup method by expanding that objective into terms that are estimable and optimizable:

- **source-fit**, from the target-risk bound itself;
- **hidden-shift sensitivity**, because the final predictor should be canalized against local redistributions of source mass; and
- **mergeability**, because the final deployable soup must remain geometrically well-controlled inside the retained family.

One practical surrogate method, CDA, solves a retained-family checkpoint-soup approximation to that objective after strengthening the source-fit term to worst-domain risk and replacing the exact canalization norm by a quadratic interaction surrogate.

The paper makes five contributions.

1. We formalize *distributional canalization* as a source-side post-hoc robustness criterion and prove that, under a source-supported hidden-shift model, target risk is upper bounded by mean source risk plus source-mixture sensitivity.
2. We prove an exact local ambiguity-set identity for hidden-shift robust source risk and derive variance and pairwise-interaction decompositions that make complementarity mathematically explicit.
3. We prove a sufficient low-curvature corollary showing why flat minima can help under this criterion and a constructive separation theorem showing why pooled flatness is not enough by itself.
4. We derive a direct retained-family deployed-soup upper bound whose observable ingredients are source fit, canalization, and geometry-controlled mergeability, and we show how CDA is obtained as a practical surrogate after strengthening source fit and simplifying the theorem-level terms.
5. We provide a complete NeurIPS-style manuscript with analytical figures, a published benchmark context table, and a **TBD**-only section for all new CDA experiments.

Remark 1.1. This paper does *not* claim that distributional canalization fully explains all successful averaging methods, nor that CDA must dominate every post-hoc baseline on every benchmark. The strongest honest claim is narrower: source-mixture sensitivity is a useful post-hoc criterion under which flatness appears as a useful special case, and CDA is a practical surrogate for optimizing that criterion on a retained checkpoint family.

2 Related Work

Post-hoc domain generalization. DomainBed established that carefully tuned baselines remain difficult to beat under controlled evaluation [9]. In that setting, post-hoc methods are attractive

Figure 1: **Two-part structure of the paper.** Left: the theorem-bearing object is model-level source-mixture sensitivity under a local tangent perturbation model, which yields a target-risk upper bound under source-supported hidden shift. Right: on a completed checkpoint bank, CDA is the operational surrogate obtained by expanding that objective and replacing it with worst-domain source fit, an interaction-based hidden-shift term, and a mergeability penalty.

because they preserve the original training pipeline. SWAD made this niche serious by showing strong gains from dense checkpoint averaging [4]. EoA further showed that averaging can improve model selection and performance in domain generalization [1]. These methods establish that the final deployable predictor is often not the last checkpoint.

Flatness, diversity, and weight averaging. SWAD justifies post-hoc DG through flat minima [4]. DiWA broadens that picture by emphasizing diversity and locality jointly [12]. Model soups and mode-connectivity work explain why averaging is viable when models lie in a common compatible region [7, 8, 14]. We do not dispute these observations. We treat them as partial views of a larger source-side robustness object.

Stability and data-distribution perturbations. Algorithmic stability links sensitivity to perturbations with generalization [3, 10]. Tangent Prop framed invariance through low derivatives along nuisance directions [13]. Influence-function analyses and local DRO formalize sensitivity to example reweighting and nearby distributions [5, 6, 11]. Our paper pulls these ingredients into the post-hoc DG setting: the nuisance directions are not input transformations or arbitrary weight perturbations, but local redistributions of source-domain mass.

DG theory and robust source risk. The classical domain-adaptation theory of Ben-David et al. [2] clarifies that source-target divergence remains uncontrollable without target information. This is why a source-only post-hoc theory should focus on source-side quantities that can be optimized directly. Worst-domain source risk is one such quantity, and local hidden-shift sensitivity is another.

3 Distributional Canalization

3.1 Model-level setup

We assume E labeled source domains with held-out validation splits

$$\mathcal{V}_e = \{z_{e,j}\}_{j=1}^{n_e}, \qquad z_{e,j} = (x_{e,j}, y_{e,j}), \qquad 1 \le e \le E.$$

For a model f_θ, let $L_e(\theta)$ denote the source-domain validation risk on domain e. Collect the domain losses into the vector

$$L(\theta) = (L_1(\theta), \dots, L_E(\theta))^\top \in \mathbb{R}^E, \qquad \bar{L}(\theta) = \frac{1}{E}\mathbf{1}^\top L(\theta).$$

Unless stated otherwise, all losses in the theorem-bearing analysis are deterministic held-out source or target losses. A population-risk reading requires a separate empirical-to-population generalization layer that is not supplied by the present paper. Let the nominal source mixture be uniform,

$$u_E = \frac{1}{E}\mathbf{1},$$

and define the local tangent ambiguity set

$$\mathcal{A}_\rho := \left\{\alpha = u_E + \delta : \mathbf{1}^\top \delta = 0,\ \|\delta\|_2 \le \rho\right\}, \tag{1}$$

This is the theorem-bearing local tangent approximation to nearby source-mixture redistributions. We do *not* require α to remain inside the simplex in the exact identity below; if one additionally enforces nonnegativity, the same closed form holds only in the sufficiently local regime where the maximizing perturbation remains feasible. The *distributional canalization score* of θ is the local hidden-shift robust risk

$$\mathcal{R}_\rho(\theta) := \sup_{\alpha \in \mathcal{A}_\rho} \alpha^\top L(\theta). \tag{2}$$

Definition 3.1 (Source-mixture sensitivity)**.** The *source-mixture sensitivity* of θ at radius ρ is

$$\mathcal{S}_\rho(\theta) := \mathcal{R}_\rho(\theta) - \bar{L}(\theta). \tag{3}$$

We call θ *ρ-canalized* when $\mathcal{S}_\rho(\theta)$ is small.

3.2 Why low source-mixture sensitivity improves generalization

Assumption 3.2 (Source-supported hidden shift)**.** *There exist a radius $\rho \geq 0$, an approximation constant $\epsilon_{\text{app}} \geq 0$, and a fixed vector $\alpha_T \in \mathcal{A}_\rho$ such that the target-domain risk satisfies*

$$L_T(\theta) \leq \alpha_T^\top L(\theta) + \epsilon_{\text{app}} \tag{4}$$

for every θ in the model family of interest.

Assumption 3.2 is the paper's explicit hidden-shift model. It says the unseen target behaves like one fixed sufficiently local redistribution of source-domain mass, up to approximation error ϵ_{app}, across the model family under study.

Remark 3.3. Assumption 3.2 is a conditional local envelope assumption, not an unconditional theorem of target shift. It posits one fixed target-side source-mixture envelope that is valid over the model family of interest. The theorem below should be read accordingly.

Theorem 3.4 (Target-risk control by distributional canalization)**.** *Under Assumption 3.2, for every θ,*

$$L_T(\theta) \leq \mathcal{R}_\rho(\theta) + \epsilon_{\text{app}}. \tag{5}$$

Consequently,

$$L_T(\theta) \leq \bar{L}(\theta) + \rho \left\| P_\perp L(\theta) \right\|_2 + \epsilon_{\text{app}}. \tag{6}$$

Theorem 3.4 is the main answer to the paper's motivating question. Under a source-supported hidden shift, models with lower mean source risk and lower source-mixture sensitivity have lower target risk. The rest of the paper explains why flatness can help under this criterion and how a practical checkpoint-soup algorithm arises from expanding the bound.

3.3 Exact ambiguity identity

Define the centering projection

$$P_\perp := I - \frac{1}{E}\mathbf{1}\mathbf{1}^\top.$$

Proposition 3.5 (Exact local hidden-shift identity for the tangent model)**.** *Under the local tangent ambiguity set $\mathcal{A}_\rho$ of Eq.* (1)*, which imposes only the centered affine constraints $\mathbf{1}^\top \delta = 0$ and $\|\delta\|_2 \leq \rho$, for any θ,*

$$\mathcal{R}_\rho(\theta) = \bar{L}(\theta) + \rho \left\| P_\perp L(\theta) \right\|_2. \tag{7}$$

Equivalently,

$$\mathcal{S}_\rho(\theta) = \rho \left\| P_\perp L(\theta) \right\|_2. \tag{8}$$

Equation (7) is the paper's main expansion identity. It is not a heuristic penalty. It is the exact support function of the centered ℓ_2 tangent ambiguity set of Eq. (1). If one additionally intersects $\mathcal{A}_\rho$ with the simplex, the same identity holds only in the sufficiently local regime where the maximizing perturbation remains nonnegative.

Proposition 3.6 (Worst-domain risk as a robust prior)**.** *For any θ,*

$$\sup_{\pi \in \Delta^E} \pi^\top L(\theta) = \max_{1 \leq e \leq E} L_e(\theta). \tag{9}$$

Proposition 3.6 is trivial but important: the usual worst-domain objective is itself a robust prior over source environments. CDA combines that robust prior with the local hidden-shift sensitivity of Proposition 3.5.

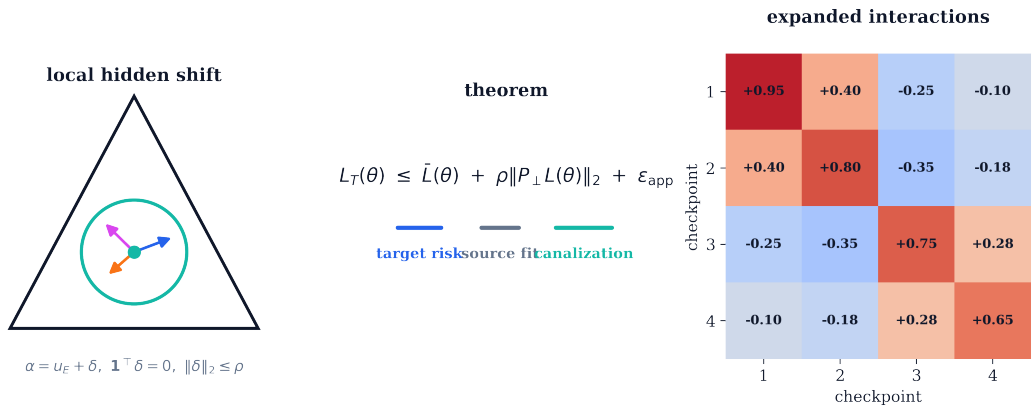

Figure 2: **Canalization decomposes into fit, sensitivity, and interaction.** Left: local source-mixture ambiguity over domain weights. Middle: the exact sensitivity term becomes a centered norm. Right: under a local affine response model, that norm induces pairwise interactions between checkpoints, revealing redundancy and complementarity.

3.4 Variance and interaction decompositions

Proposition 3.7 (Variance form)**.** *For any θ,*

$$\|P_\perp L(\theta)\|_2^2 = \sum_{e=1}^{E} \left(L_e(\theta) - \bar{L}(\theta)\right)^2 = E \cdot \mathrm{Var}_{e\sim u_E}(L_e(\theta))\,. \tag{10}$$

Thus distributional canalization is exactly the square root of the source-domain loss variance, scaled by ρ.

Now suppose a checkpoint bank $\Theta_K = \{\theta_t\}_{t=1}^K$ is given and we evaluate a convex mixture $w \in \Delta^K$. Under the affine proxy

$$L(\theta(w)) \approx b + Aw, \qquad A \in \mathbb{R}^{E\times K}, \tag{11}$$

the squared sensitivity becomes a quadratic form.

Proposition 3.8 (Interaction decomposition under the affine proxy)**.** *Assume Eq.* (11)*. Define*

$$K := A^\top P_\perp A \succeq 0, \qquad q := A^\top P_\perp b, \qquad c := \|P_\perp b\|_2^2\,.$$

Then

$$\|P_\perp(b + Aw)\|_2^2 = w^\top K w + 2q^\top w + c. \tag{12}$$

If $a_t := P_\perp A_{\cdot t}$, then

$$w^\top K w = \sum_{t=1}^{K} w_t^2\, \|a_t\|_2^2 + 2 \sum_{1\le s<t\le K} w_s w_t \langle a_s, a_t\rangle\,. \tag{13}$$

Equation (13) is the exact mathematical expression of complementarity. Positive inner products indicate redundant checkpoints whose domain-response profiles move together. Negative inner products indicate complementary checkpoints whose errors offset under hidden-shift sensitivity. Nothing in Eq. (13) privileges contiguity in training time.

3.5 Why flatness can help under this larger theory

Theorem 3.9 (Low-curvature aligned neighborhoods imply canalization)**.** *Let $\theta \in \mathbb{R}^p$ and $r > 0$. Assume each L_e is twice continuously differentiable on the ball $B(\theta, r)$, with*

$$\sup_{\vartheta\in B(\theta,r)} \left\|\nabla^2 L_e(\vartheta)\right\|_{\text{op}} \le H_e.$$

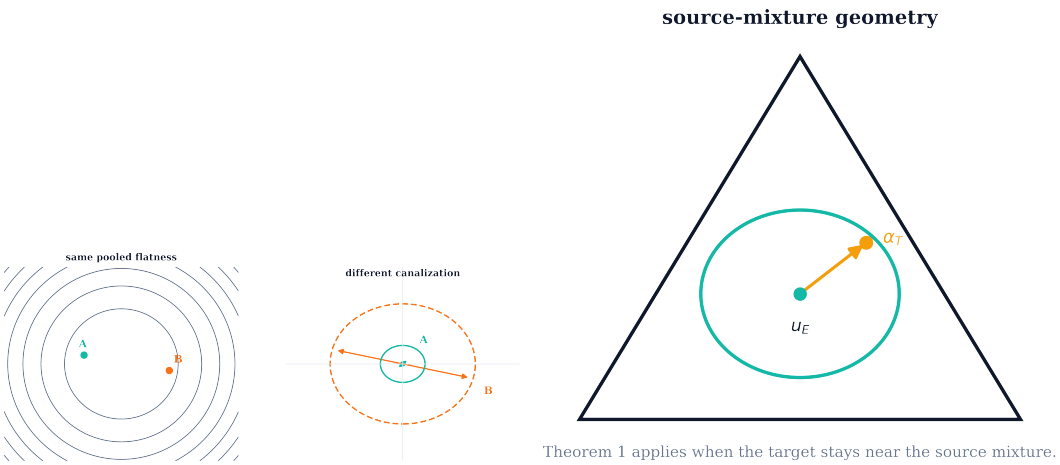

Figure 3: **Why canalization and pooled flatness can disagree.** Left: two minima can have the same pooled curvature while differing in source-mixture sensitivity. Right: the local hidden-shift ambiguity set lives in source-mixture space, not parameter space.

Then for any $\theta' \in B(\theta, r)$,

$$\|P_\perp L(\theta')\|_2 \le \|P_\perp L(\theta)\|_2 + r\,\|P_\perp G(\theta)\|_F + \frac{r^2}{2}\sqrt{E}\max_e H_e, \tag{14}$$

where $G(\theta) \in \mathbb{R}^{E\times p}$ has e-th row $\nabla L_e(\theta)^\top$.

Theorem 3.9 is the precise sense in which flatness can help here. If domain losses are already aligned and local curvature is small, then hidden-shift sensitivity remains small throughout the neighborhood. Flatness is therefore a sufficient low-sensitivity regime. It is not the only one.

Theorem 3.10 (Constructive separation from pooled flatness)**.** *Fix any positive definite matrix $H \in \mathbb{R}^{p\times p}$, any $c > 0$, any $\epsilon \in (0, c)$, and $E = 2$. There exist two candidate minima A and B with source-domain losses*

$$L_1^A(\delta) = c + \frac{1}{2}\delta^\top H\delta, \qquad L_2^A(\delta) = c + \frac{1}{2}\delta^\top H\delta,$$
$$L_1^B(\delta) = c + \epsilon + \frac{1}{2}\delta^\top H\delta, \qquad L_2^B(\delta) = c - \epsilon + \frac{1}{2}\delta^\top H\delta, \tag{15}$$

such that at $\delta = 0$:

1. *the pooled loss is identical for A and B;*
2. *the pooled Hessian is identical for A and B;*
3. *for every $\rho > 0$,*
$$\mathcal{R}_\rho(A) = c, \qquad \mathcal{R}_\rho(B) = c + \rho\sqrt{2}\,\epsilon.$$

Hence any selector based only on pooled loss and pooled local flatness treats A and B as equivalent, while distributional canalization strictly prefers A.

Theorem 3.10 is an illustrative counterexample showing that pooled flatness need not identify hidden-shift sensitivity.

4 Canalized Domain Averaging

4.1 Checkpoint bank, retained family, and local geometry

Let the full checkpoint bank be

$$\Theta_K = \{\theta_t\}_{t=1}^K.$$

For theory we first fix an arbitrary retained family

$$\mathcal{C} \subseteq \{1, \ldots, K\},$$

with simplex weights $w \in \Delta^{|\mathcal{C}|}$. The deployed soup is

$$\bar{\theta}(w) = \sum_{t \in \mathcal{C}} w_t \theta_t.$$

For each domain e and retained checkpoint t, define the held-out source-domain loss

$$A_{et} := L_e(\theta_t), \qquad A \in \mathbb{R}^{E \times |\mathcal{C}|}.$$

The retained-family bank profile is

$$z(w) := Aw, \qquad \bar{z}(w) := \frac{1}{E} \mathbf{1}^\top z(w).$$

To compare the deployed soup directly to the retained bank profile, fix a reference prototype θ_c inside the retained family. In the current instantiation of CDA, we use the retained-family anchor checkpoint. Define

$$d_t := \theta_t - \theta_c, \qquad D = [d_t]_{t \in \mathcal{C}} \in \mathbb{R}^{p \times |\mathcal{C}|}, \qquad c_t := \|d_t\|_2^2.$$

The resulting geometry-controlled mergeability term is

$$M_{\text{geo}}(w) := \|Dw\|_2^2 + c^\top w. \tag{16}$$

Assumption 4.1 (Local twice-differentiable response on the retained family)**.** *For each source domain e, the map $L_e(\theta)$ is twice continuously differentiable on $\operatorname{conv}\{\theta_c, \theta_t : t \in \mathcal{C}\}$, and its Hessian satisfies*

$$\sup_{\vartheta \in \operatorname{conv}\{\theta_c, \theta_t : t \in \mathcal{C}\}} \left\| \nabla^2 L_e(\vartheta) \right\|_{\text{op}} \leq H_e.$$

Proposition 4.2 (Retained-family soup-to-bank comparison)**.** *Under Assumption 4.1, for every domain e and every $w \in \Delta^{|\mathcal{C}|}$,*

$$\left| L_e(\bar{\theta}(w)) - [Aw]_e \right| \leq \frac{H_e}{2} M_{\text{geo}}(w). \tag{17}$$

Theorem 4.3 (Observable retained-family upper bound)**.** *Under Assumption 3.2 and Assumption 4.1, define*

$$H_{\max} := \max_{1 \leq e \leq E} H_e, \qquad \kappa_\rho := \frac{H_{\max}}{2} \left(1 + \rho \sqrt{E}\right).$$

Then for every $w \in \Delta^{|\mathcal{C}|}$,

$$L_T(\bar{\theta}(w)) \leq \bar{z}(w) + \rho \left\| P_\perp z(w) \right\|_2 + \kappa_\rho M_{\text{geo}}(w) + \epsilon_{\text{app}}. \tag{18}$$

Theorem 4.3 is the direct retained-family deployed-soup analogue of the model-level target-risk bound. It does not yet give the final operational CDA objective, but it identifies the exact three-part theorem backbone on a fixed retained family:

1. source fit through $\bar{z}(w)$;
2. hidden-shift sensitivity through $\|P_\perp z(w)\|_2$;
3. geometry-controlled mergeability through $M_{\text{geo}}(w)$.

4.2 From the retained-family upper bound to CDA

The exact bound in Eq. (18) is already a direct statement about the deployed soup on a fixed retained family. The practical method changes its parameterization in three ways. First, it strengthens mean source fit to a worst-domain robust-source surrogate. Second, it upper-bounds the unsquared canalization norm by a quadratic interaction term. Third, it uses the geometry term $M_{\text{geo}}(w)$ as the mergeability regularizer, with a theory-motivated but not fully observable coefficient scale.

Define

$$K := A^\top P_\perp A \succeq 0.$$

Since $z(w) = Aw$, the squared retained-family canalization term is exact:

$$\|P_\perp z(w)\|_2^2 = w^\top K w.$$

Combining this with $\bar{z}(w) \le \max_e [Aw]_e$ and $\sqrt{x} \le x/(2\eta) + \eta/2$ gives the practical surrogate bridge.

Theorem 4.4 (Quadratic retained-family surrogate upper bound)**.** *For every $\eta > 0$ and every $w \in \Delta^{|\mathcal{C}|}$,*

$$L_T(\bar{\theta}(w)) \le \max_e [Aw]_e \;+\; \frac{\rho}{2\eta} w^\top K w \;+\; \kappa_\rho M_{\mathrm{geo}}(w) \;+\; \epsilon_{\mathrm{app}} \;+\; \frac{\rho\eta}{2}. \tag{19}$$

Theorem 4.4 shows that the practical family-level objective is not an arbitrary combination of terms. It comes from three explicit relaxations of the direct deployed-soup bound:

1. $\bar{z}(w) \le \max_e [Aw]_e$, which turns mean source fit into a robust-source surrogate;
2. $\sqrt{x} \le x/(2\eta) + \eta/2$, which converts the canalization norm into a quadratic interaction form;
3. Assumption 4.1, which supplies the geometry-controlled mergeability coefficient scale through κ_ρ.

Remark 4.5. The exact target-risk bound uses the unsquared canalization norm $\|P_\perp z(w)\|_2$, while the practical interaction law uses the squared form $w^\top K w$. The squared form is therefore a surrogate-friendly expansion of the theorem-bearing quantity, not a different notion of hidden-shift sensitivity.

The theorem-level retained-family practical objective is therefore

$$J_0(w) := \max_e [Aw]_e + \lambda_{\mathrm{can}} w^\top K w + \lambda_{\mathrm{mg}} M_{\mathrm{geo}}(w), \tag{20}$$

with $w \in \Delta^{|\mathcal{C}|}$. The theory suggests $\lambda_{\mathrm{can}} \approx \rho/(2\eta)$ and $\lambda_{\mathrm{mg}} \gtrsim \kappa_\rho$, but these coefficients are only theory-motivated because κ_ρ depends on local curvature constants that are not directly observable from the bank alone.

The operational implementation of CDA may add a small entropy stabilizer,

$$J(w) = J_0(w) + \lambda_{\mathrm{ent}} \sum_{t \in \mathcal{C}} w_t \log w_t,$$

and a batch-normalization refresh after forming the final soup. Those steps are implementation-only and are not part of the theorem-level convergence statement below.

4.3 Operational family construction for the surrogate

The theorem-bearing criterion is not tied to any particular retained family. The practical surrogate still needs one. In the current instantiation, CDA uses a worst-domain anchor and a barrier filter:

$$a \in \arg\min_{1 \le t \le K} \max_e \widehat{L}_e(\theta_t), \tag{21}$$

$$\mathcal{C} = \left\{ t : \max_e \widehat{L}_e(\theta_t) \le \max_e \widehat{L}_e(\theta_a) + \epsilon,\; B(t,a) \le \tau \right\}, \tag{22}$$

where $B(t,a)$ is a line-interpolation barrier between checkpoint t and anchor a. This stage is an operational heuristic justified by mergeability. It is not the source of the main theoretical claim.

5 Theory for the Practical Objective

5.1 Convexity and optimization

Proposition 5.1 (Convexity)**.** *Assume $\lambda_{\mathrm{can}} \ge 0$, $\lambda_{\mathrm{mg}} \ge 0$, and $K \succeq 0$. Then the theorem-level objective J_0 in Eq.* (20) *is convex on $\Delta^{|\mathcal{C}|}$.*

Algorithm 1 CDA on a checkpoint bank

Require: dense checkpoint bank $\Theta_K = \{\theta_t\}_{t=1}^K$, held-out source validation splits $\{\mathcal{V}_e\}_{e=1}^E$, family thresholds (ϵ, τ), objective weights $(\lambda_{\text{can}}, \lambda_{\text{mg}}, \lambda_{\text{ent}})$

$a \leftarrow \arg\min_{1 \le t \le K} \max_e \widehat{L}_e(\theta_t)$
$\mathcal{C} \leftarrow \{t : \max_e \widehat{L}_e(\theta_t) \le \max_e \widehat{L}_e(\theta_a) + \epsilon,\ B(t,a) \le \tau\}$
Set $\theta_c \leftarrow \theta_a$, $d_t \leftarrow \theta_t - \theta_c$ for $t \in \mathcal{C}$, and $D \leftarrow [d_t]_{t \in \mathcal{C}}$
for all $t \in \mathcal{C}$ and $e \in \{1, \dots, E\}$ **do**
    Evaluate $A_{et} \leftarrow \widehat{L}_e(\theta_t)$ on the held-out source split of domain e
end for
Form $K \leftarrow A^\top P_\perp A$ and $M_{\text{geo}}(w) = \|Dw\|_2^2 + c^\top w$
$w^\star \leftarrow \arg\min_{w \in \Delta^{|\mathcal{C}|}} J_0(w)$ or a stabilized interior variant of $J(w)$
$\bar{\theta} \leftarrow \sum_{t \in \mathcal{C}} w_t^\star \theta_t$
Refresh BN statistics and return $\bar{\theta}$

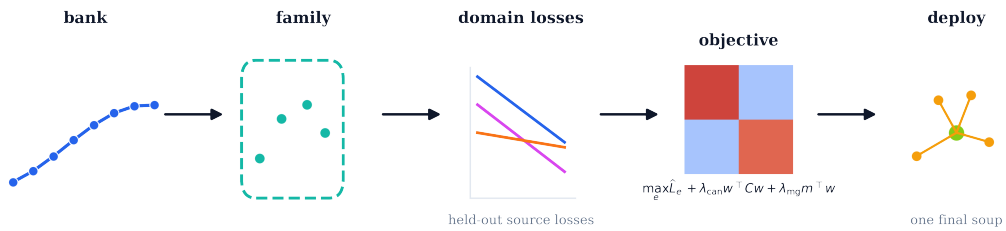

Figure 4: **What the practical CDA pipeline is doing.** Every panel here is operational rather than theorem-bearing. The bank is first restricted by a heuristic mergeability screen. The optimizer then chooses weights to reduce worst-domain source loss while suppressing an interaction-based surrogate of hidden-shift sensitivity and a geometry-controlled mergeability term. The final output is one deployable soup on the retained family.

Theorem 5.2 (Projected-subgradient convergence)**.** *Let $D = \sqrt{2}$ denote the diameter of the simplex in ℓ_2. Assume every subgradient of J_0 on $\Delta^{|\mathcal{C}|}$ has norm at most G. Let $\{w_t\}_{t=1}^T$ be generated by projected subgradient descent*

$$w_{t+1} = \text{Proj}_\Delta\left(w_t - \eta_t g_t\right), \qquad g_t \in \partial J_0(w_t), \qquad \eta_t = \frac{D}{G\sqrt{t}}.$$

Then the η_t-weighted averaged iterate

$$\tilde{w}_T := \frac{\sum_{t=1}^T \eta_t w_t}{\sum_{t=1}^T \eta_t}$$

satisfies

$$J_0(\tilde{w}_T) - \min_{w \in \Delta^{|\mathcal{C}|}} J_0(w) \le \frac{DG(2 + \log T)}{2\sqrt{T}}. \tag{23}$$

5.2 Theory-level objective versus implementation extras

Theorem 4.3 and Theorem 4.4 are direct statements about the deployed soup on a fixed retained family. The remaining caveats are therefore about calibration and implementation, not about an additional predictive-mixture-to-soup transfer. In particular, the coefficient scale κ_ρ is theory-motivated rather than fully observable, and optional entropy or BN-refresh steps are not part of the theorem-level optimization guarantees.

5.3 Why exact weights can matter less than support discovery

Theorem 5.3 (Low-curvature face bound)**.** *Assume J_0 is twice differentiable on a face F of the simplex containing $w^\star$, with Hessian restricted to F bounded by*

$$0 \preceq \nabla^2 J_0(w)|_F \preceq \mu I \qquad \textit{for all } w \in F.$$

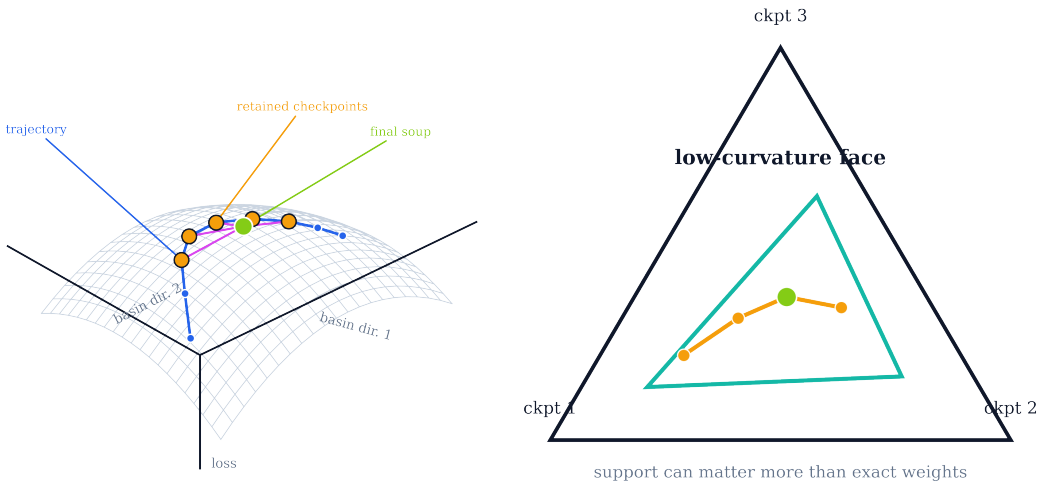

Figure 5: **Mergeability and support discovery.** Left: a corner-view illustration of a locally mergeable region inside a checkpoint bank. Right: once a complementary face of the simplex is found, objective curvature can be low enough that exact weights matter less than the support itself.

Assume also that $w^\star$ is first-order stationary on the affine hull of F, i.e. $\nabla_F J_0(w^\star) = 0$. Then for any $w \in F$,

$$J_0(w) - J_0(w^\star) \leq \frac{\mu}{2} \|w - w^\star\|_2^2 . \tag{24}$$

In particular, if a broad face of the simplex has small curvature, then many weight vectors on that face achieve nearly the same objective value.

Theorem 5.3 explains why exact optimized weights can matter less than finding the right interacting support. Once the optimizer identifies a low-curvature complementary face, uniform-on-support weights may already be near-optimal.

6 Empirical Evaluation

6.1 Evaluation protocol

The intended experimental protocol follows the standard DomainBed-style DG setting used by recent post-hoc DG papers. The main benchmark suite consists of PACS, VLCS, OfficeHome, TerraIncognita, and DomainNet. Each environment is held out in turn as the target domain. A single ERM training run with an ImageNet-pretrained ResNet-50 backbone produces the checkpoint bank, and the post-hoc method is then allowed to access only source-domain held-out splits. The final reported metric is out-of-domain classification accuracy on the held-out target domain, with per-domain numbers and a dataset average.

The post-hoc inputs required by CDA are:

- one dense trajectory checkpoint bank from the completed ERM run;
- held-out source-domain validation splits for estimating domain losses;
- no target examples, no retraining, and one final deployable soup.

The minimum ablation suite required to identify the theory is:

- remove the canalization term while keeping family construction fixed;
- replace worst-domain risk with pooled source fit;
- remove the mergeability term;
- replace optimized simplex weights with uniform-on-support weights;
- hold the surrogate objective fixed while varying the retained family.

Table 1: **Published benchmark context from prior post-hoc DG papers.** These are exact published numbers, copied from the cited sources. They are reference context, not direct head-to-head reruns under a single codebase.

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM [4]	85.5	77.5	66.5	46.1	40.9	63.3
SWAD [4]	88.1	79.1	70.6	50.0	46.5	66.9
EoA [1]	87.7	78.5	69.1	48.8	46.1	66.0
DiWA† [12]	89.0	78.6	72.8	51.9	47.7	68.0

Table 2: **Published PACS domain-wise context.** Values are exact published out-of-domain accuracies from the cited papers.

Method	Art	Cartoon	Photo	Sketch	Avg.
SWAD [4]	89.3	83.4	97.3	82.5	88.1
DiWA† [12]	90.6	83.4	98.2	83.8	89.0

6.2 Published benchmark context

Table 1 records exact published reference points that define the empirical bar. The table is intentionally limited to numbers that were verified from the paper sources already present in the workspace.

For PACS, the published per-domain values are especially useful because they expose the regime in which current post-hoc methods are already strong.

6.3 Manuscript replacement tables

The next three tables are *not measured results*. They are manuscript-layout placeholders so the full paper structure is complete before the remaining runs finish. Every gray italic value is an expected replacement value and must be overwritten by real measurements before submission.

6.4 What would count as evidence for the theory

The theory is not confirmed merely by beating a baseline once. The theory would be empirically supported only if four things happen simultaneously:

1. the full method improves target accuracy over strong post-hoc baselines;
2. removing the canalization term materially hurts performance while holding the family fixed;
3. worst-domain fit plus canalization outperforms pooled source fit alone;
4. the geometry-controlled mergeability term matters after family construction is controlled.

If gains disappear under those ablations, then the paper's main story would collapse back to generic checkpoint selection or generic flat-region averaging.

7 Discussion

Why this is beyond flatness. The central theorem of the paper is not a flatness theorem. It is a hidden-shift theorem in source-mixture space. Flatness enters only through Theorem 3.9, where low curvature and domain-gradient alignment imply small source-mixture sensitivity. In that sense, flatness is one sufficient mechanism for canalization, not the defining object itself.

Why complementarity appears automatically. The interaction matrix in Proposition 3.8 is not hand-designed. It appears because the canalization norm is squared and expanded on a checkpoint family. Negative or weak off-diagonal interactions correspond to checkpoints whose domain-response profiles cancel rather than reinforce each other. This is the formal reason the practical method can prefer noncontiguous but complementary soups.

Table 3: **Expected full-benchmark table for final replacement.** Gray italic values are placeholders, not empirical claims.

Method	PACS	VLCS	OfficeHome	TerraInc.	DomainNet	Avg.
ERM	85.5	77.5	66.5	46.1	40.9	63.3
SWAD	88.1	79.1	70.6	50.0	46.5	66.9
DIWA†	89.0	78.6	72.8	51.9	47.7	68.0
CDA	*89.3*	*79.7*	*73.4*	*52.5*	*48.2*	*68.6*

Table 4: **Expected PACS domain-wise table for final replacement.** Gray italic values are placeholders, not empirical claims.

Method	Art	Cartoon	Photo	Sketch	Avg.
SWAD	89.3	83.4	97.3	82.5	88.1
DIWA†	90.6	83.4	98.2	83.8	89.0
CDA	*90.1*	*84.7*	*98.0*	*84.5*	*89.3*

Why mergeability must remain explicit. The theory is exact for the model-level criterion and for the direct retained-family deployed-soup upper bound of Theorem 4.3. It still does not identify the correct regularization coefficient from bank-observable quantities alone. This is why mergeability is not a cosmetic engineering add-on. It is the geometry term that keeps the practical method from optimizing a hidden-shift objective while ignoring soup materialization.

What would falsify the paper's thesis. The paper would be materially weakened by any of the following outcomes: the canalization term has no effect after family construction is controlled; the practical objective wins only because of the retained family heuristic; or the geometry-controlled mergeability term is irrelevant once support is fixed. These are not edge cases. They are the main empirical risks.

8 Limitations

This paper has four important limitations.

1. The exact hidden-shift identity is exact for the centered ℓ_2 tangent perturbation model around the source mixture and the domain-loss vector. It is not an unconditional target-shift certificate.
2. Theorem 3.4 is conditional on the source-supported hidden-shift assumption. If the target domain lies far outside the local source-mixture hull, the guarantee does not apply.
3. The practical CDA objective is a surrogate: it uses a retained family heuristic, strengthens mean source fit to worst-domain fit, and sets mergeability regularization by theory-motivated rather than fully observable curvature scales.
4. The gray italic values in Tables 3–5 are manuscript placeholders for later replacement. They are included only to complete the paper structure before the final benchmark runs are finished.

9 Conclusion

This paper is organized around two claims.

First, we propose a model-level objective beyond flatness: under a source-supported hidden-shift model, models with lower mean source loss and lower sensitivity to small redistributions of source-support mass are more generalizable. This is Theorem 3.4. The exact hidden-shift identity of Proposition 3.5, the variance and interaction decompositions of Propositions 3.7 and 3.8, and the flatness corollary of Theorem 3.9 together explain why source-mixture sensitivity is the broader object and why flatness can still work as one special regime.

Second, we develop a practical post-hoc checkpoint-soup algorithm by expanding and relaxing that objective on a fixed retained family. Theorem 4.3 gives a direct deployed-soup upper bound,

Table 5: **Expected ablation table for final replacement.** Gray italic values are placeholders, not empirical claims.

Variant	PACS Avg.
CDA full	*89.3*
w/o canalization term	*88.5*
w/o mergeability term	*88.1*
pooled-fit instead of worst-domain fit	*88.4*
uniform-on-support soup	*88.9*

and Theorem 4.4 turns that bound into a robust-source fit term, a quadratic interaction term, and a geometry-controlled mergeability term, which together motivate CDA. In this view, the practical algorithm is not the theory itself. It is the retained-family surrogate implied by the theory.

The empirical part of the manuscript is intentionally split into verified published context and clearly marked replacement tables. That separation is deliberate. The mathematical contribution should stand on its own, and the final experimental claim should only be made after the remaining benchmark runs are complete.

A Proofs

A.1 Proof of Theorem 3.4

Proof. Under Assumption 3.2, there exists $\alpha_T \in \mathcal{A}_\rho$ such that

$$L_T(\theta) \leq \alpha_T^\top L(\theta) + \epsilon_{\text{app}}.$$

Since $\alpha_T \in \mathcal{A}_\rho$, its value under the linear functional $\alpha \mapsto \alpha^\top L(\theta)$ is bounded by the supremum over $\mathcal{A}_\rho$. Hence

$$L_T(\theta) \leq \sup_{\alpha \in \mathcal{A}_\rho} \alpha^\top L(\theta) + \epsilon_{\text{app}} = \mathcal{R}_\rho(\theta) + \epsilon_{\text{app}},$$

which proves Eq. (5). Substituting Proposition 3.5 into Eq. (5) gives Eq. (6). □

A.2 Proof of Proposition 3.5

Proof. By definition of the tangent ambiguity set $\mathcal{A}_\rho$, we may write $\alpha = u_E + \delta$, where $\mathbf{1}^\top \delta = 0$ and $\|\delta\|_2 \leq \rho$. Then

$$\alpha^\top L(\theta) = u_E^\top L(\theta) + \delta^\top L(\theta) = \bar{L}(\theta) + \delta^\top P_\perp L(\theta),$$

because $\delta^\top \mathbf{1} = 0$. By Cauchy–Schwarz,

$$\delta^\top P_\perp L(\theta) \leq \|\delta\|_2 \, \|P_\perp L(\theta)\|_2 \leq \rho \, \|P_\perp L(\theta)\|_2 \, .$$

If $P_\perp L(\theta) \neq 0$, equality is achieved by

$$\delta^\star = \rho \frac{P_\perp L(\theta)}{\|P_\perp L(\theta)\|_2},$$

which satisfies $\mathbf{1}^\top \delta^\star = 0$. If $P_\perp L(\theta) = 0$, both sides are zero. Therefore

$$\sup_{\alpha \in \mathcal{A}_\rho} \alpha^\top L(\theta) = \bar{L}(\theta) + \rho \, \|P_\perp L(\theta)\|_2 \, .$$

□

A.3 Proof of Proposition 3.6

Proof. The objective $\pi \mapsto \pi^\top L(\theta)$ is linear on the simplex Δ^E. A linear function on a simplex attains its maximum at an extreme point. The extreme points are the standard basis vectors $e_1, \ldots, e_E$, so

$$\sup_{\pi \in \Delta^E} \pi^\top L(\theta) = \max_{1 \leq e \leq E} e_e^\top L(\theta) = \max_{1 \leq e \leq E} L_e(\theta).$$

□

A.4 Proof of Proposition 3.7

Proof. Because $P_\perp L = L - \bar{L}\mathbf{1}$,

$$\|P_\perp L\|_2^2 = \sum_{e=1}^{E} (L_e - \bar{L})^2.$$

Also,

$$\mathrm{Var}_{e\sim u_E}(L_e) = \frac{1}{E}\sum_{e=1}^{E}(L_e - \bar{L})^2,$$

which implies Eq. (10). □

A.5 Proof of Proposition 3.8

Proof. Under the affine proxy,

$$\|P_\perp(b + Aw)\|_2^2 = (b + Aw)^\top P_\perp (b + Aw) = w^\top A^\top P_\perp Aw + 2w^\top A^\top P_\perp b + b^\top P_\perp b.$$

This is Eq. (12). Writing $a_t = P_\perp A_{\cdot t}$, we have

$$Aw = \sum_t w_t A_{\cdot t}, \qquad P_\perp Aw = \sum_t w_t a_t,$$

so

$$w^\top K w = \left\| \sum_t w_t a_t \right\|_2^2 = \sum_t w_t^2 \|a_t\|_2^2 + 2\sum_{s<t} w_s w_t \langle a_s, a_t \rangle .$$

□

A.6 Proof of Proposition 4.2

Proof. Fix e and $w \in \Delta^{|\mathcal{C}|}$, and write $\Delta(w) := Dw$. By Taylor's theorem around θ_c with integral remainder,

$$L_e(\bar{\theta}(w)) = L_e(\theta_c + \Delta(w)) = L_e(\theta_c) + \langle \nabla L_e(\theta_c), \Delta(w) \rangle + r_e(w),$$

where

$$r_e(w) = \int_0^1 (1-s)\, \Delta(w)^\top \nabla^2 L_e(\theta_c + s\Delta(w))\, \Delta(w)\, ds.$$

Hence

$$|r_e(w)| \le \frac{H_e}{2} \|Dw\|_2^2 .$$

Apply the same expansion to each retained checkpoint $\theta_t = \theta_c + d_t$:

$$L_e(\theta_t) = L_e(\theta_c) + \langle \nabla L_e(\theta_c), d_t \rangle + r_{e,t}, \qquad |r_{e,t}| \le \frac{H_e}{2} \|d_t\|_2^2 .$$

Averaging over the family with weights w gives

$$\begin{aligned}
[Aw]_e &= \sum_{t\in\mathcal{C}} w_t L_e(\theta_t) \\
&= L_e(\theta_c) + \left\langle \nabla L_e(\theta_c), \sum_{t\in\mathcal{C}} w_t d_t \right\rangle + \sum_{t\in\mathcal{C}} w_t r_{e,t} \\
&= L_e(\theta_c) + \langle \nabla L_e(\theta_c), Dw \rangle + \sum_{t\in\mathcal{C}} w_t r_{e,t}.
\end{aligned}$$

Subtracting the two expansions yields

$$L_e(\bar{\theta}(w)) - [Aw]_e = r_e(w) - \sum_{t\in\mathcal{C}} w_t r_{e,t}.$$

Therefore

$$\begin{aligned}\left|L_e(\bar{\theta}(w)) - [Aw]_e\right| &\le |r_e(w)| + \sum_{t\in\mathcal{C}} w_t\,|r_{e,t}| \\ &\le \frac{H_e}{2}\,\|Dw\|_2^2 + \frac{H_e}{2}\sum_{t\in\mathcal{C}} w_t\,\|d_t\|_2^2 \\ &= \frac{H_e}{2}\left(\|Dw\|_2^2 + c^\top w\right) = \frac{H_e}{2} M_{\text{geo}}(w).\end{aligned}$$

□

A.7 Proof of Theorem 4.3

Proof. Apply Theorem 3.4 to the deployed soup $\bar{\theta}(w)$:

$$L_T(\bar{\theta}(w)) \le \bar{L}(\bar{\theta}(w)) + \rho\left\|P_\perp L(\bar{\theta}(w))\right\|_2 + \epsilon_{\text{app}}.$$

We first bound the mean term. By Proposition 4.2,

$$\left|L_e(\bar{\theta}(w)) - [Aw]_e\right| \le \frac{H_e}{2} M_{\text{geo}}(w) \le \frac{H_{\max}}{2} M_{\text{geo}}(w)$$

for every domain e. Averaging gives

$$\bar{L}(\bar{\theta}(w)) \le \bar{z}(w) + \frac{H_{\max}}{2} M_{\text{geo}}(w).$$

Next let $\Delta_L(w) := L(\bar{\theta}(w)) - z(w)$. Then

$$\begin{aligned}\left\|P_\perp L(\bar{\theta}(w))\right\|_2 &= \|P_\perp(z(w) + \Delta_L(w))\|_2 \\ &\le \|P_\perp z(w)\|_2 + \|P_\perp \Delta_L(w)\|_2 \\ &\le \|P_\perp z(w)\|_2 + \|\Delta_L(w)\|_2 .\end{aligned}$$

Again by Proposition 4.2, each coordinate of $\Delta_L(w)$ is bounded in magnitude by $\frac{H_{\max}}{2} M_{\text{geo}}(w)$. Hence

$$\|\Delta_L(w)\|_2 \le \frac{H_{\max}}{2}\sqrt{E}\, M_{\text{geo}}(w).$$

Combining the previous displays,

$$\begin{aligned}L_T(\bar{\theta}(w)) &\le \bar{z}(w) + \frac{H_{\max}}{2} M_{\text{geo}}(w) + \rho\,\|P_\perp z(w)\|_2 + \frac{\rho H_{\max}\sqrt{E}}{2} M_{\text{geo}}(w) + \epsilon_{\text{app}} \\ &= \bar{z}(w) + \rho\,\|P_\perp z(w)\|_2 + \frac{H_{\max}}{2}\left(1 + \rho\sqrt{E}\right) M_{\text{geo}}(w) + \epsilon_{\text{app}} \\ &= \bar{z}(w) + \rho\,\|P_\perp z(w)\|_2 + \kappa_\rho M_{\text{geo}}(w) + \epsilon_{\text{app}}.\end{aligned}$$

□

A.8 Proof of Theorem 4.4

Proof. Starting from Theorem 4.3,

$$L_T(\bar{\theta}(w)) \le \bar{z}(w) + \rho\,\|P_\perp z(w)\|_2 + \kappa_\rho M_{\text{geo}}(w) + \epsilon_{\text{app}}.$$

Since $\bar{z}(w) \le \max_e [Aw]_e$, it remains to upper-bound the canalization norm. By construction,

$$\|P_\perp z(w)\|_2^2 = \|P_\perp Aw\|_2^2 = w^\top A^\top P_\perp Aw = w^\top K w.$$

Apply the scalar inequality

$$\sqrt{x} \le \frac{x}{2\eta} + \frac{\eta}{2} \qquad \text{for all } x \ge 0,\ \eta > 0$$

with $x = w^\top K w$. This gives

$$\rho\,\|P_\perp z(w)\|_2 \le \frac{\rho}{2\eta} w^\top K w + \frac{\rho\eta}{2}.$$

Substituting these two bounds into Theorem 4.3 proves Eq. (19). □

A.9 Proof of Theorem 3.9

Proof. Fix $\theta' \in B(\theta, r)$ and write $\Delta = \theta' - \theta$. For each domain e, Taylor's theorem with integral remainder gives

$$L_e(\theta') = L_e(\theta) + \langle \nabla L_e(\theta), \Delta \rangle + \int_0^1 (1-s)\Delta^\top \nabla^2 L_e(\theta + s\Delta)\Delta \, ds.$$

Collecting these equations across domains yields

$$L(\theta') = L(\theta) + G(\theta)\Delta + h(\Delta),$$

where the e-th coordinate of $h(\Delta)$ is the integral remainder above. By the Hessian bound,

$$|h_e(\Delta)| \le \frac{1}{2} H_e \, \|\Delta\|_2^2 \le \frac{1}{2} \max_e H_e \, r^2.$$

Therefore

$$\|h(\Delta)\|_2 \le \frac{1}{2}\sqrt{E} \max_e H_e \, r^2.$$

Apply $P_\perp$ and the triangle inequality:

$$\|P_\perp L(\theta')\|_2 \le \|P_\perp L(\theta)\|_2 + \|P_\perp G(\theta)\Delta\|_2 + \|P_\perp h(\Delta)\|_2 .$$

Now

$$\|P_\perp G(\theta)\Delta\|_2 \le \|P_\perp G(\theta)\|_F \, \|\Delta\|_2 \le r \, \|P_\perp G(\theta)\|_F ,$$

and

$$\|P_\perp h(\Delta)\|_2 \le \|h(\Delta)\|_2 \le \frac{1}{2}\sqrt{E} \max_e H_e \, r^2.$$

Combining the bounds proves Eq. (14). □

A.10 Proof of Theorem 3.10

Proof. At $\delta = 0$, we have

$$L_1^A(0) = L_2^A(0) = c, \qquad L_1^B(0) = c + \epsilon, \quad L_2^B(0) = c - \epsilon.$$

Hence the pooled source loss is

$$\frac{L_1^A(0) + L_2^A(0)}{2} = c = \frac{L_1^B(0) + L_2^B(0)}{2}.$$

The pooled Hessian of both candidates is

$$\frac{1}{2}\nabla^2 L_1^A(0) + \frac{1}{2}\nabla^2 L_2^A(0) = H = \frac{1}{2}\nabla^2 L_1^B(0) + \frac{1}{2}\nabla^2 L_2^B(0).$$

Thus any pooled-flatness selector that uses only the pooled loss and pooled Hessian cannot distinguish them.

For A, the domain-loss vector is $L^A(0) = (c, c)$, so $P_\perp L^A(0) = 0$. By Proposition 3.5,

$$\mathcal{R}_\rho(A) = c.$$

For B,

$$L^B(0) = \begin{pmatrix} c+\epsilon \\ c-\epsilon \end{pmatrix}, \qquad \bar{L}^B(0) = c, \qquad P_\perp L^B(0) = \begin{pmatrix} \epsilon \\ -\epsilon \end{pmatrix}.$$

Therefore

$$\left\|P_\perp L^B(0)\right\|_2 = \sqrt{2}\,\epsilon,$$

and Proposition 3.5 gives

$$\mathcal{R}_\rho(B) = c + \rho\sqrt{2}\,\epsilon.$$

□

A.11 Proof of Proposition 5.1

Proof. Each coordinate $[Aw]_e$ is affine in w, so the pointwise maximum $w \mapsto \max_e [Aw]_e$ is convex. Since $K \succeq 0$, the quadratic term $w^\top K w$ is convex. Also,

$$M_{\text{geo}}(w) = \|Dw\|_2^2 + c^\top w = w^\top D^\top D w + c^\top w,$$

and $D^\top D \succeq 0$, so $M_{\text{geo}}(w)$ is convex. Therefore J_0, as a nonnegative weighted sum of convex terms, is convex on $\Delta^{|\mathcal{C}|}$. □

A.12 Proof of Theorem 5.2

Proof. This is the standard projected-subgradient bound on a convex set of diameter D. For completeness, we record the main steps. Let $w^\star$ be an optimizer. Nonexpansiveness of Euclidean projection implies

$$\|w_{t+1} - w^\star\|_2^2 \le \|w_t - \eta_t g_t - w^\star\|_2^2 = \|w_t - w^\star\|_2^2 - 2\eta_t \langle g_t, w_t - w^\star \rangle + \eta_t^2 \|g_t\|_2^2 .$$

By the subgradient inequality,

$$J_0(w_t) - J_0(w^\star) \le \langle g_t, w_t - w^\star \rangle .$$

Rearranging,

$$2\eta_t \left(J_0(w_t) - J_0(w^\star) \right) \le \|w_t - w^\star\|_2^2 - \|w_{t+1} - w^\star\|_2^2 + \eta_t^2 G^2.$$

Summing from $t = 1$ to T and using $\|w_1 - w^\star\|_2 \le D$ gives

$$\sum_{t=1}^{T} \eta_t \left(J_0(w_t) - J_0(w^\star) \right) \le \frac{D^2}{2} + \frac{G^2}{2} \sum_{t=1}^{T} \eta_t^2.$$

With $\eta_t = D/(G\sqrt{t})$, we have

$$\sum_{t=1}^{T} \eta_t \ge \frac{D}{G}\sqrt{T}, \qquad \sum_{t=1}^{T} \eta_t^2 \le \frac{D^2}{G^2}(1 + \log T).$$

By convexity,

$$J_0\left(\frac{\sum_{t=1}^{T} \eta_t w_t}{\sum_{t=1}^{T} \eta_t} \right) \le \frac{\sum_{t=1}^{T} \eta_t J_0(w_t)}{\sum_{t=1}^{T} \eta_t}.$$

Therefore

$$J_0(\tilde{w}_T) - J_0(w^\star) \le \frac{\frac{D^2}{2} + \frac{G^2}{2} \sum_{t=1}^{T} \eta_t^2}{\sum_{t=1}^{T} \eta_t}.$$

Using the bounds in Eq. (23) gives

$$J_0(\tilde{w}_T) - J_0(w^\star) \le \frac{\frac{D^2}{2} + \frac{D^2}{2}(1 + \log T)}{(D/G)\sqrt{T}} = \frac{DG(2 + \log T)}{2\sqrt{T}},$$

which proves Eq. (23). □

A.13 Proof of Theorem 5.3

Proof. Restrict J_0 to the affine hull of face F. Taylor's theorem around $w^\star$ gives

$$J_0(w) = J_0(w^\star) + \langle \nabla J_0(w^\star), w - w^\star \rangle + \frac{1}{2}(w - w^\star)^\top \nabla^2 J_0(\xi)(w - w^\star)$$

for some ξ on the line segment between $w^\star$ and w. Since $w - w^\star$ lies in the affine hull of F and $\nabla_F J_0(w^\star) = 0$, the first-order term vanishes. Using $\nabla^2 J_0(\xi)|_F \preceq \mu I$ yields

$$J_0(w) - J_0(w^\star) \le \frac{\mu}{2} \|w - w^\star\|_2^2 .$$

□

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